



THE UNIVERSITY OF
MELBOURNE

SWEN90016

Software Processes and Management

Semester 1, 2026 Lecture 4

*Faculty of Engineering and Information Technology
School of Computing and Information Systems*

The University of Melbourne acknowledges the Traditional Owners of the unceded land on which we work, learn and live: the Wurundjeri Woi-wurrung and Bunurong peoples (Burnley, Fishermans Bend, Parkville, Southbank and Werribee campuses), the Yorta Yorta Nation (Dookie and Shepparton campuses), and the Dja Dja Wurrung people (Creswick campus).

The University also acknowledges and is grateful to the Traditional Owners, Elders and Knowledge Holders of all Indigenous nations and clans who have been instrumental in our reconciliation journey.

We recognise the unique place held by Aboriginal and Torres Strait Islander peoples as the original owners and custodians of the lands and waterways across the Australian continent, with histories of continuous connection dating back more than 60,000 years. We also acknowledge their enduring cultural practices of caring for Country.

We pay respect to Elders past, present and future, and acknowledge the importance of Indigenous knowledge in the Academy. As a community of researchers, teachers, professional staff and students we are privileged to work and learn every day with Indigenous colleagues and partners.

Please ask your questions about the SWEN90016 Lecture

Nobody has responded yet.

Hang tight! Responses are coming in.

Project Management Plan – Formal



Almost every organisations will have its own “*version*” of a Project Management Plan (PMP), however the reasons they have and use them are the same.

A PMP is a formal approved document that defines how the project is executed, monitored and controlled. It may be a summary or a detailed document.

It is a document that is owned, controlled and populated by the Project Manager and is used throughout the project.

A good PMP provides the required level of detail across key project components and is the one source of truth for all parties involved across the project.

[PROJECT INTEGRATION MANAGEMENT \[Readings Online in CANVAS\]](#)



Project Management Plan

Project Charter V Project Management Plan



A Project Charter is a summary project proposal to secure approval for the project goals and terms (useful as part of Business Case).

A PMP is an approved document showing how to achieve the approved project goals / benefits and provides the details on how to execute and manage the project (used as part of mobilisation and on-going management of the project).

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Primary Use: Summary (few pages) of key information used to communicate, engage, gain buy-in and obtain approvals.

Project Management Plan - Template				
1	2	3	4	5
1	Project Purpose	Project Objectives	Project Overview	Project Schedule
2	Project Objectives	Project Overview	Project Schedule	Project Charter Approval
3	Project Overview	Project Schedule	Project Charter Approval	Appendix A: References
4	Project Schedule	Project Charter Approval	Appendix A: References	Appendix B: Key Terms
5	Project Charter Approval	Appendix A: References	Appendix B: Key Terms	
6	Appendix A: References	Appendix B: Key Terms		
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Primary Use: Detailed document used to establish and manage the project. Defines all key items the project needs to consider.



Project Management Plan



A typical PMP (Formal) consists of all / or most of the following categories.

- *Project Information*
 - Executive Summary
 - Financial Authority to proceed
 - Key Stakeholders
 - Scope
 - Delivery approach / SDLC – e.g. Waterfall or Agile
 - Resources / People
 - Key Milestones
 - Project Budget
 - Lessons learned applied to this project
 - Constraints

Project Management Plan



And.....

- *Project Governance*
 - Roles and Responsibilities
 - Mandatory Project Planning / Key Additional Activities
 - Schedule
 - Risk Management
 - Cost Estimation
 - Quality Assurance
 - Configuration Management (Change Management)

The PMP is a large multi-page document that takes time to prepare, review and complete. Multiple people (subject experts) are involved and prepare the specific details. The Project Manager coordinates all items and has ultimate accountability for the quality and outcome.

Project Management Plan – Sample



Refer to CANVAS – Week 4 Lecture Modules for sample PMPs

IMPORTANT – The PMP document is only a **SAMPLE Project Management Plan**

PMP is not a uniform template but rather a customizable framework that organizations tailor to their specific project needs. This flexibility allows organizations to align project execution with established standards such as PMBOK or Agile methodologies while accommodating unique project requirements.

For SWEN90016, a custom PMP document has been created.

Agile: To document or not to document?

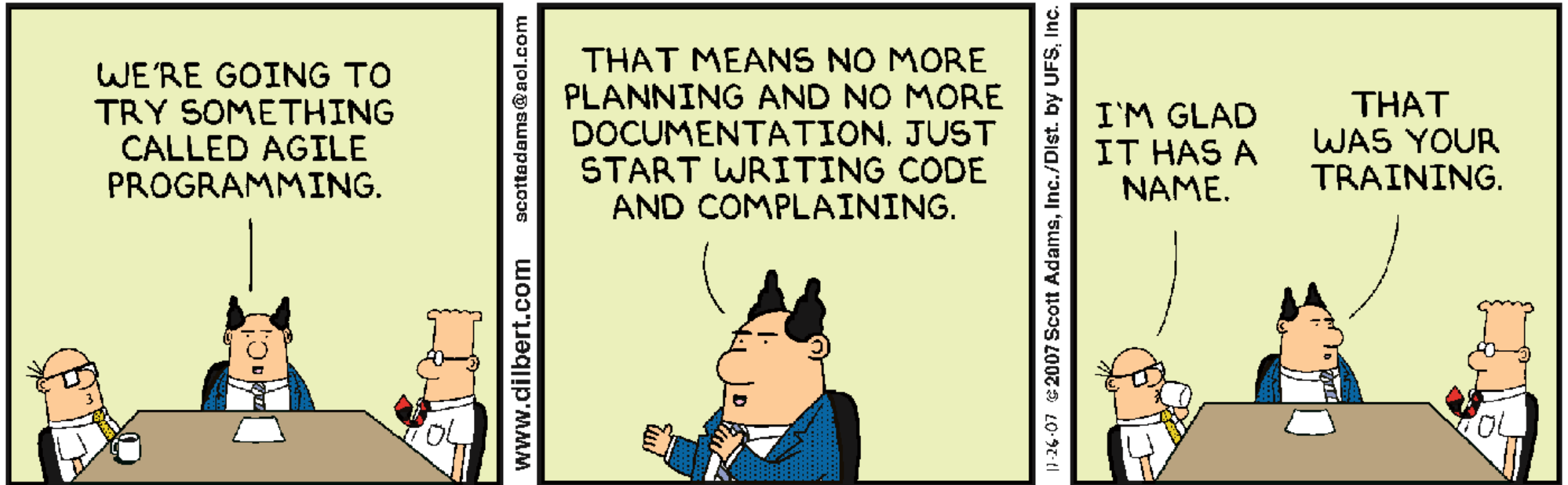


Image Courtesy: [Scrum Alliance Twitter Handle Post](#)

Agile: To document or not to document?



Myth:

1. Agile Means No Documentation
2. Agile Development is often misunderstood as eliminating the need for documentation

Reality:

1. Agile values working software over comprehensive documentation, but does not dismiss documentation entirely
2. This misconception stems from the Agile Manifesto's emphasis on working software, leading some to believe that documentation is unnecessary

Agile: To document or not to document?



Agile Encourages Smart Documentation:

There are project situations in which documentation is absolutely required

– User Stories, Flowcharts, Wireframes, Client Meetings, Regulatory Requirements

Agile suggests to be “SMART”^[1] about the Documentation

1. Agile promotes 'just barely good enough' (JBGE) documentation—sufficient to support the project without overburdening the team.
2. Documentation is created 'just in time' (JIT) when it is needed, not upfront, to avoid outdated or unnecessary details.
3. The focus is on living, collaborative documentation that evolves with the project.

[1] Specific, Measurable, Attainable, Relevant, and Time-bound

[Just Barely Good Enough \(JBGE\) Artifacts: An Agile Core Practice](#)

[Lean/Agile Documentation: Strategies for Agile Teams](#)



Why Document in Agile?



1. To meet stakeholder requirements and maintain organizational memory.
 - While Working Software is the primary goal, you also need to develop supporting documentation required to use, operate, support, maintain, and extend the software over time
2. To support communication with external groups, such as software outsourcing teams.
 - Not always possible to always have all stakeholders available
 - Not always possible to co-locate a development team
3. For audit and compliance purposes.
 - Depending on the type of system that you are developing, there may be some audit needs. In these cases, you need to follow a well-defined process and capture proof that you did so
4. To present, clarify, and solidify complex ideas.

What to Document in Agile?



Examples of Documents that you may create during your project: (Not Exhaustive)

Product Vision:

- Essence of the product, including cost estimates, predicted benefits, risks, and milestones.

Project Overview:

- Summary of critical project information like user contacts, technologies, and tools.
- Maintained throughout development for onboarding new team members.

Design Decisions:

- Record of key design and architecture choices made during the project.
- Integrated into system overviews and source code.

Requirements Documents:

- Overview of system functionality through use cases, user stories, or prototypes.
- Captures requirements as executable specifications.

Operations Documentation:

- Describes system dependencies, backup procedures, and troubleshooting guidelines.

System Documentation:

- Technical and business architecture overview, covering high-level requirements.
- Ensures critical information is preserved if the development team transitions.
- Designed to minimize the need for extensive user training.

What to Document in Agile?

And of course.....Considering Scrum SDLC



Scrum Artifacts:

- Information that a Scrum Team and stakeholders use to describe the product being developed, actions undertaken to produce the same, and actions performed during the execution of the project.
- Essential for every Scrum Team to enable transparency, inspection, and adaption.

Product Vision, Product Backlog

Sprint Vision, Sprint Backlog

Product Increments

Burndown Charts

Definition of Done

[The 7 Scrum Artifacts: Definitions and Examples](#)





Stakeholder Management



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Please ask your questions about the SWEN90016 Lecture

Nobody has responded yet.

Hang tight! Responses are coming in.

Scenario 1 – Dashboard Demonstration



- **You are demonstrating the updated software to your client at the end of a Sprint.**
- **The client picks up on one element of the system behaviour**
 - The client says that the observed behaviour is a bug.
 - The client describes the behaviour they expected to see.
- **You are confident that the observed behaviour is what the client requested.**
- **You even remember the relevant interaction with the client.**

How do you respond to the Client in the demonstration?

How do you respond to this client during the software demonstration?

Nobody has responded yet.

Hang tight! Responses are coming in.

Scenario 2: Workflow Change – New System Feature



A new software system that you are developing for a client introduces a different workflow for approvals

- 1. In the old system, approvals were sent on emails to be approved manually.**
- 2. In the new system, approvals are done using a central dashboard with automated reminders for the same.**
- 3. Users are complaining – "The new system is more complicated than the old way"**

(Reality – New Workflow reduces errors and speeds up the approval process)

How would you respond to end user complaints ?

How would you respond to user complaints (Workflow Change)?

Nobody has responded yet.

Hang tight! Responses are coming in.

Please ask your questions about the SWEN90016 Lecture

Nobody has responded yet.

Hang tight! Responses are coming in.

What is Stakeholder Analysis?



Stakeholder Identification: Identifying all individuals, groups, or organizations who may be affected by, or have an interest in the project, whether directly or indirectly.

Understanding Stakeholder Interests: Examining stakeholders' interests, objectives, expectations, and concerns regarding the project to ensure their needs are addressed.

Assessing Stakeholder Influence: Evaluate the level of influence or power each stakeholder holds over the project's outcomes, decision-making processes, or resources.

Prioritizing Stakeholder Engagement: Prioritize stakeholders based on their importance, level of influence, or degree of impact on the project's success.

Managing Stakeholder Relationships: Proactively manage relationships with stakeholders by anticipating their reactions, addressing potential conflicts, and fostering open communication and collaboration.

Stakeholder management



Identifying Stakeholders & the Stakeholder Register

Internal Stakeholders	External Stakeholders
Shareholders	End Users / Customers
Employees	Suppliers
Board Members	Governments
Sponsor / Business Managers	Unions
Project Manager	Local Communities / General Public
Management	Other Related Institutions
Project Team	Competitors

Stakeholder management



Identifying Stakeholders & the Stakeholder Register

Name	Position	Internal/External	Project Role	Contact Information
Stephen	VP of Operations	Internal	Project Sponsor	stephen@globaloil.com
Betsy	CFO	Internal	Senior Manager. Approves Funds	betsy@globaloil.com
Chien	CIO	Internal	Senior Manager. PM's Boss	chien@globaloil.com
Ryan	IT Analyst	Internal	Team Member	ryan@globaloil.com
Lori	Director Accounting	Internal	Senior Manager	lori@globaloil.com
Sanjay	Director Refineries	Internal	Senior Manager of Largest Refinery	sanjay@globaloil.com
Debra	Consultant	External	Project Manager	debra@globaloil.com
Suppliers	Suppliers	External	Software Supplier	suppliers@gmail.com

[Project Stakeholder Management \[Readings Online in CANVAS\]](#)



Stakeholder Engagement and Planning



Levels of Stakeholder Engagement

Unaware: Unaware of the project and its potential impacts on them

Resistant: Aware of the project yet resistant to change

Neutral: Aware of the project yet neither supportive nor resistant

Supportive: Aware of the project and supportive of change

Champion / Leading: Aware of the project and drives change

[Project Stakeholder Management \[Readings Online in CANVAS\]](#)



Stakeholder Engagement and Planning



The stakeholder management plan can include:

- Current and desired engagement levels
- Interrelationships between stakeholders
- Communication requirements
- Potential management strategies for each stakeholder
- Methods for updating the stakeholder management plan

Note: Because a stakeholder management plan often includes sensitive information, it may not be part of the official project documents, which are normally available for all stakeholders to review.

In many cases, only project managers and a few other team members should prepare the stakeholder management plan. Parts of the stakeholder management plan may not be written down, and if they are, distribution is limited.

[Project Stakeholder Management \[Readings Online in CANVAS\]](#)



Stakeholder Engagement and Planning



Stakeholder Analysis includes:

- Names and Organisations of Key Stakeholders
- Their Role on the Project
- Unique Facts about Each Stakeholder
- Level of Interest in the Project
- Influence on the Project
- Suggestions and Strategies for Managing Relationships with each Stakeholder

Stakeholder Engagement and Planning



Understanding Stakeholder Engagement and Planning

Stakeholder Analysis example:

Name	Power / Influence	Current Engagement	Potential Management Strategies
Brian	High/High	Champion / Leading	Brian can seem intimidating due to his physical stature and deep voice, but he has a great personality and sense of humour. He previously led a similar software upgrade project at another company and knows what he wants. Manage closely and ask for his advice as required. He likes to be kept in touch with short, frequent updates in person.
Mary	High/Medium	Resistant	Mary is very organised yet hardhead. She has been pushing corporate IT standards, and the system the PM and sponsors like best goes against those standards, even though it's the best solution for this project and the company as a whole. Need to convince her that it is ok and that people still respect her work and position.
Finance Team	Medium/High	Resistant	The Finance Team is resistant to the Project. They believe the funds can be better used elsewhere in the organisations. They also believe the Benefits are not achievable in the defined pay back period. Key stakeholder as they control the funding which the project is dependant on. Need to convince them of the costs & benefits and ensure they understand the detail. Get them involved in a detailed review of all \$'s.
Jessica	High/Low	Neutral	Very professional, logical person. Gets along with Mary. She has supported Brian in approving past projects with strong business cases. Provide detailed financial justification for the suggested solution to keep her satisfied. Also ask her to talk to Mary on Brian's behalf.

[Project Stakeholder Management \[Readings Online in CANVAS\]](#)



Case Study – Formal



You are developing a new e-commerce platform for a retail clothing store.

Who are the stakeholders???

Who are the stakeholders of this project?

Nobody has responded yet.

Hang tight! Responses are coming in.

Case Study – Formal



You are developing a new e-commerce platform for a retail clothing store.

Who are the stakeholders???

- Internal:
 - IT Department (Project manager, developers, testers)
 - Marketing Department (Campaign manager)
 - Sales Department (Sales representatives)
 - Management (CEO, CFO)
- External:
 - Payment processing company
 - Web hosting provider

Case Study – Agile

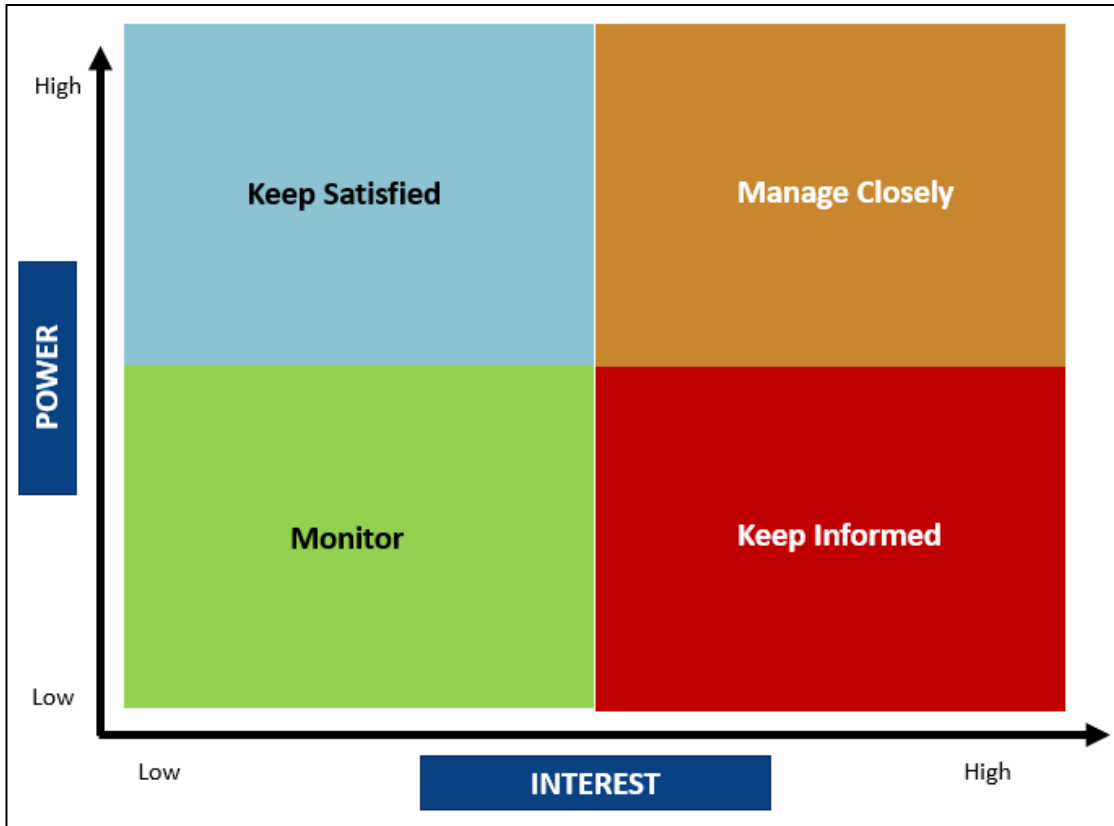


You are developing a mobile application for a food delivery service.

Who are the stakeholders???

- Internal:
 - Product Owner (Represents stakeholders' needs)
 - Developers
 - Scrum Master (Facilitates communication)
- External:
 - Customers (App users)
 - Restaurant partners

Understanding the Power-Interest Grid



- The Power Interest Grid is a popular tool used for stakeholder analysis.
- By understanding stakeholders' power and interest levels, organizations can tailor their strategies to build positive relationships and achieve desired outcomes.
- Categorizes stakeholders based on their level of power and their level of interest in the project.



Understanding the Power Interest Grid



High Power, Low Interest Quadrant

- Stakeholders in this quadrant possess significant power but have a low level of interest in the project, but their decisions can significantly impact the project.
- Examples include senior executives who delegate project responsibilities, financial backers who are primarily concerned with returns on investment, and government agencies providing regulatory oversight.
- Strategies could involve periodic updates, providing high-level summaries, and ensuring their needs are addressed efficiently to maintain their support.

Low Power, Low Interest Quadrant

- Stakeholders in this quadrant have minimal power and interest in the project.
- While they may not require extensive engagement, overlooking their concerns entirely could still result in reputational risks or unintended consequences.
- Strategies could involve minimal communication, monitoring for any emerging issues, and addressing concerns to maintain overall project support and credibility.

High Power, High Interest Quadrant

- Stakeholders in this quadrant have significant power and a high level of interest in the project.
- They can heavily influence project outcomes and decisions.
- Examples include key decision-makers, major investors, regulatory bodies, and influential community leaders.
- Strategies could involve active engagement, frequent communication, and seeking their input and feedback to ensure their needs and concerns are addressed.

Low Power, High Interest Quadrant

- Stakeholders in this quadrant have a high level of interest in the project but limited power to influence outcomes directly.
- While they may not hold significant sway over decisions, their support or opposition can impact public perception and project acceptance.
- Strategies may involve engaging them in consultation, addressing their concerns, and leveraging their support to influence stakeholders with higher power.

Teams and Groups



- It is included in Engineers Australia Stage 1 competencies
 - Teamwork is huge part of Australian work culture.
 - Most software products are too large for a single programmer (in available time)
 - If software components are written by different people from different background, the software will have higher chance to meets the needs of bigger demographic
 - You argue but also learn from each other.
-
- A group is a collection of people
 - A team is a group that is (or is expected to be)
 - cohesive
 - focused on a common purpose or goal
 - All teams are groups, but not all groups are teams!



Why Teams?



- The ability to work with others is an important life and career skill, no matter what your future plans are
 - almost all nontrivial software development is done as a team
 - almost all nontrivial research is done as a team
 - surveys of potential employers of IT graduates consistently rate teamwork and interpersonal skills
- Software engineering is about delivering systems that are large or complicated or mission-critical (or all of these things!)
 - To make these systems work, we will need
 - lots of people
 - working together effectively
 - And these teams need to be structured carefully, because the structure of the organization can affect the success of the project

Group work isn't popular with students

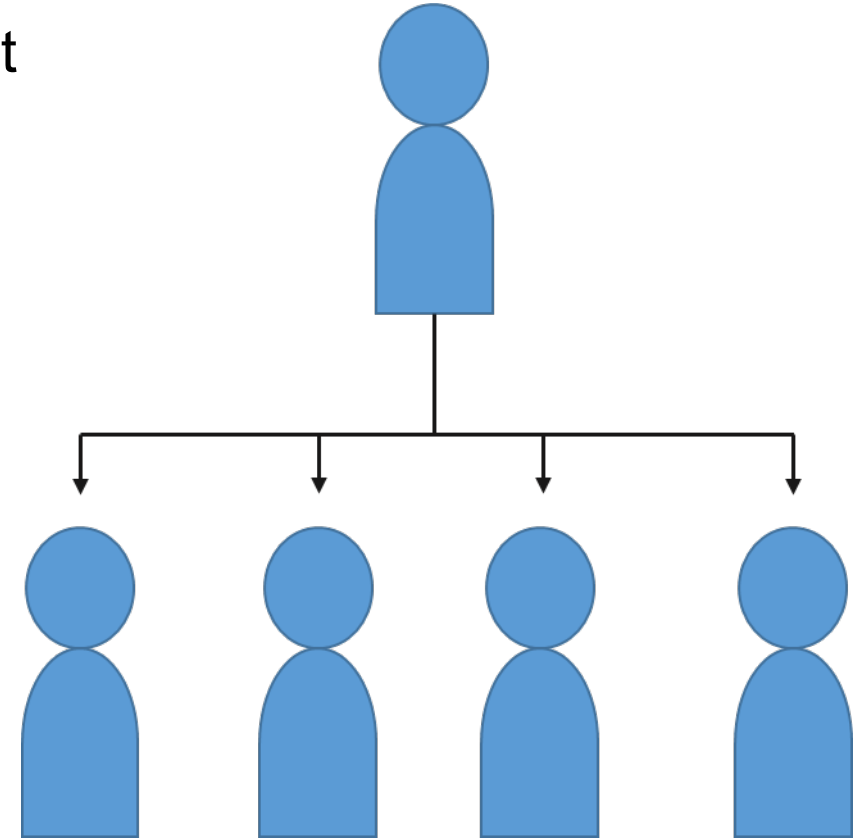


- Many students dread working on a project with others
 - coordination can be difficult
 - workload can be unevenly divided
 - teammates' work might not be good enough, or not completed on time (or at all)
- But most of these problems can also arise in industry
 - in SWEN90016 we are trying to address them the same way they'd be addressed in the workplace
 - risk management, task/time tracking, informal peer evaluations
- Sometimes students fear group work because they are worried about how their group will be led/managed
 - don't want to be bossed around

Why teams can be terrible: Autocracy

Sometimes students think this is what teams should be like.
The boss hands out orders, team members carry them out
This is a bad team structure!

- little opportunity for teammates to collaborate
- morale tends to suffer
- boss is a single point of failure – if they get sick or are transferred, team cannot function

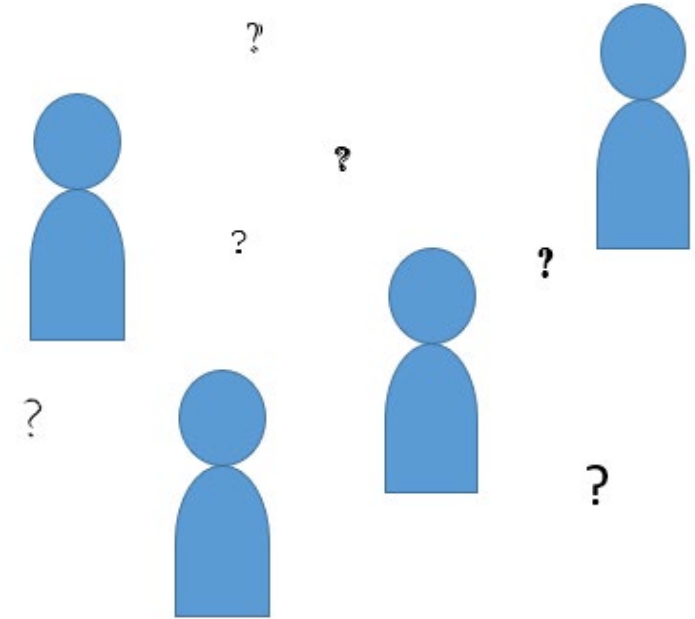


Why teams can be terrible: Anarchy



This can arise if teams fear autocracy but don't know what else they can do.

- Good news: no boss!
- Bad news: nobody knows what to do!



Why teams can be terrible: Democratic



Sounds like a simple idea: make all decisions collaboratively

One problem is that “all decisions” might turn out to be a lot

- team gets bogged down in decision-making
- little gets achieved

Another problem: dilution of individual responsibility

- if anything goes wrong, it's nobody's fault
- so less incentive to make good decisions



Which team is better?

Democratic team

Autocratic team



When your
boss asks for
proof you're
in hospital



Collaborative Teams

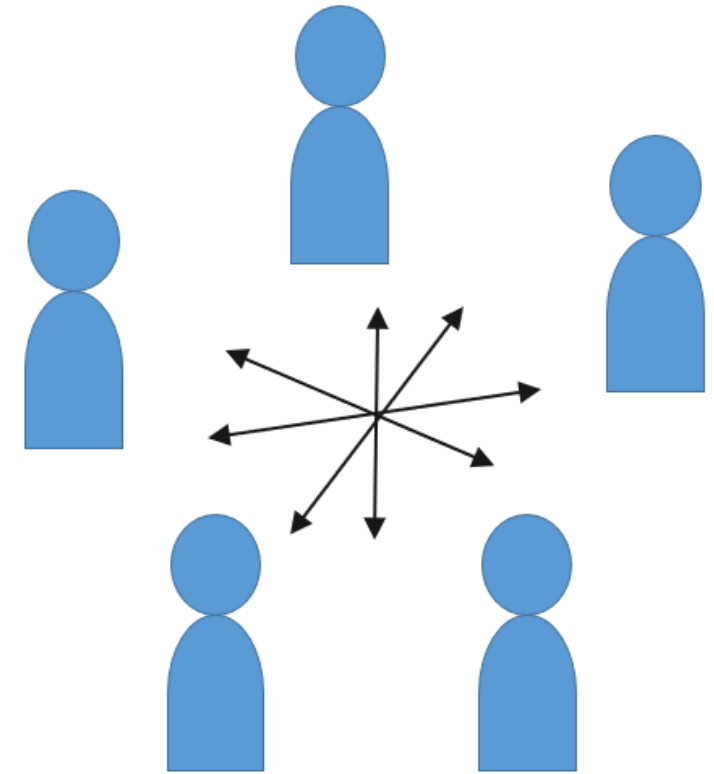
Effective teams tend to have a fairly flat structure

- team members may have different responsibilities

Members make decisions within their area of expertise

- so all teammates participate in decision making to some extent
- avoids the problems of autocracy
- avoids the problems of democracy
 - don't have to consult the whole team for every decision
 - only those with relevant expertise

Agile teams are collaborative in nature



Extreme Programming Teams



- All code is written by two programmers sharing a computer.
 - “Pair programming”
 - Driver: write the code
 - Navigator: check or review the code
- Test cases drawn up by one member of team, but testing done by the other member.
- Knowledge not all lost if one programmer leaves
- Inexperienced and experienced programmers can learn from each other.



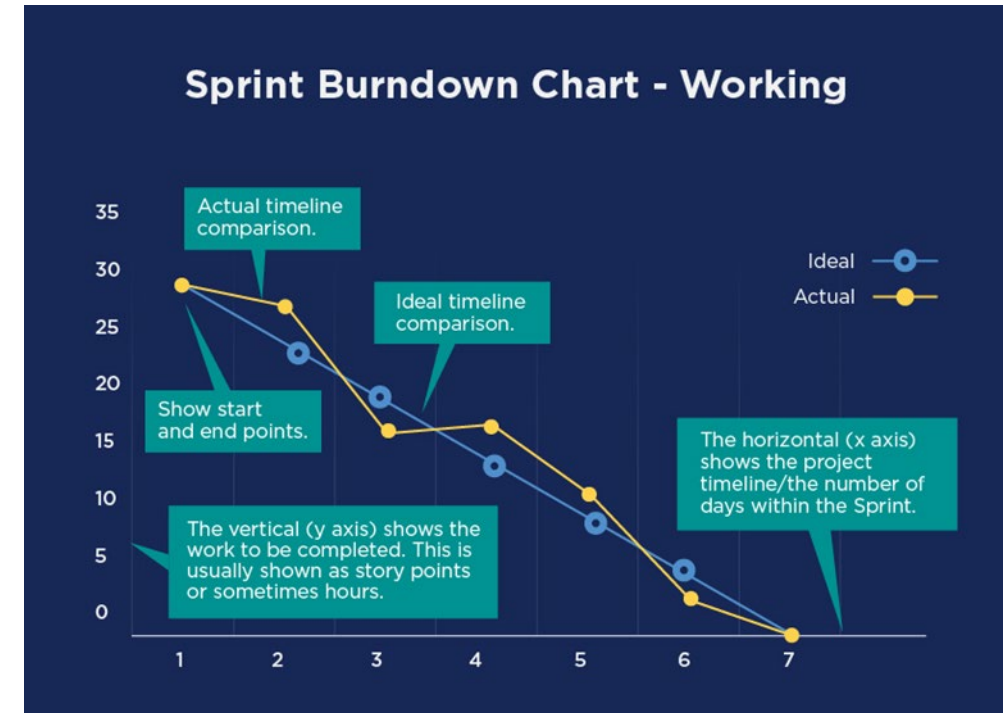
Scrum Teams



- A variant of collaborative team model and a bit democratic as well.
- Team size of 7 +/- 2
- Accountabilities
- “Scrum Master”
 - Ensures team is following rules of Scrum.
 - Protects team from outside barriers.
- “Product Owner”
 - Representative of client/customer
 - ONE person.
 - Responsible for what will be developed and in what order.
- “Developers”
 - Responsible for determining how to deliver and to some extent what’s been decided in the Sprint Backlog and by the Product Owner.

Scrum Teams

- fixed-length “iterations”
- Team agrees how much work to be done in an iteration
- Team members sign up for tasks within an iteration (Sprint – 2-4 weeks in length preferably).
- Number of mechanisms
 - Customer/Client sees progress at the end of every Sprint!
 - Everybody sees what everybody else is doing
 - Burndown charts track progress within Sprint
 - It shows how many tasks still need to be done in a Sprint.



How do you make a great Team?



- In 1987, Tom DeMarco and Timothy Lister wrote a book called Peopleware about management in IT and engineering
- Tried to write a chapter about how to ensure that a team would work well
 - they failed to do this...
- But were able to write a chapter on how to ensure that a team won't work well!
 - they identified some bad practices which they called *teamicide*

What is Teamicide?



- Defensive management
 - if management makes it clear to the team that they're not trusted, the team loses motivation
- Bureaucracy
 - hard to maintain your enthusiasm when you're doing lots of mindless paperwork
- Physical separation
 - teams work best when members have spontaneous, casual interactions as well as planned and guided interactions
 - spontaneous interactions enable creativity but are impossible if team members aren't co-located
- Fragmentation of time
 - because you can't concentrate if you keep switching back and forth between projects

What is Teamicide?



- Quality-reduced product
 - idea might be to lower costs or to ship earlier but it's still demotivating to ship a poor product
- Phony deadlines
 - i.e. trying to make team members work faster by imposing artificially-close deadlines
 - really a form of defensive management: management doesn't trust the team to work hard unless they are nagged
- Clique control
 - i.e. management may have policies that actively interfere with the health of teams (e.g. regularly rotating people through different roles)
 - the authors see this as a consequence of management's failure to understand teams, since teams seldom exist at management level

If you have a conflict in your team, what should be the "First Step" towards resolving the conflict?

0%



Attack each other verbally

0%



Discuss the problems and reasoning causing the conflict

0%



Inform your mentor and ask for suggestions

0%



Inform your mentor and ask them to resolve the conflict

0%



Go behind each others back and try to get a majority vote

Please ask your questions about the SWEN90016 Lecture

Nobody has responded yet.

Hang tight! Responses are coming in.

Distributed Teams



Team distributed across geographic locations, relying on communication technology.

- Aka virtual team^[1], geographically dispersed team, remote team

Why?

- Support region, live in region, work from home, access to high-level or lower-cost talent

Comms:

- Less frequent and less rich (visual and behavioural clues)
- Can introduce cultural or other overheads/barriers (e.g. timezones)
- Can reduce biases, help comms with local stakeholders, support for 24-hour work cycle
- Visual team activities (e.g. design) can be particularly challenging



Communication Management



Why are we discussing communications management in a software processes and management course?

ACS Accreditation Requirement for the course

Graduate Employability one of the most important aims of higher education in most OECD countries ^[1]



Question Emoji Vectors
by Vecteezy

[1] The undergraduate self-perception of employability: human capital, careers advice, and career ownership
[Readings Online in CANVAS]



Communication Management



Why are we discussing communications management in a software processes and management course?

Technical and discipline competencies are important, but employers place a premium on graduate employability skills. The two pillars of employability skills are **Teamwork and Communication** ^{[2][3][12]}

Among the 14 key employability skills ordered by priority and importance and expected to be seen and fully developed in candidates^[4], **Communication Skills** is ranked at 1, and **Teamwork** is ranked at 4.

^[2] [Employers' perceptions of the employability skills of new graduates \[Readings Online in CANVAS\]](#)

^[3] [Re-engineering graduate skills – a case study \[Readings Online in CANVAS\]](#)

^[4] [Employability skills explored: shedding light on the views of stakeholders and employers \[Readings Online in CANVAS\]](#)

^[12] [Graduate Employability of Business Students \[Readings Online in CANVAS\]](#)



[Question Emoji Vectors by Vecteezy](#)



Communication Management

Communication Skills and Team Work is one of the five pillars of the core competency model to assess the employability of graduates and improve their outcomes^[5]

The two common themes across existing employability frameworks from education sectors of UK, USA and Australia are - **Communication Skills and Teamwork**^[6]

Among the six "Personal and Social Capabilities" that are crucial for employability, **Communication Skills** is ranked at 2^[10]

Verbal and Non-verbal components of **Communication Skills** play a significant role in the employability of engineering students^[13].

^[5] [An integrative model for measuring graduates' employability skills—A study in China \[Readings Online in CANVAS\]](#)

^[6] [The student perspective on employability \[Readings Online in CANVAS\]](#)

^[10] [Employability skills: what makes us employable? \[Readings Online in CANVAS\]](#)

^[13] [Evaluating the Role of the Communication Skills of Engineering Students on Employability According to the Outcome-Based Education \(OBE\) Theory \[Readings Online in CANVAS\]](#)



Question Emoji Vectors
by Vecteezy



BUT.....



"**Effective Communication**" - one of the most important skills area, and one that graduates are thought to lack^[7].

A recent international survey found that less than half of employers believed that new graduates "are adequately prepared"^[8]

Being able to **communicate** clearly is a necessity in almost every professional position, but "**an attribute graduates often lack**"^[9]



Emoji icons created by Vectors
Market - Flaticon

[7] The myth of job readiness? Written communication, employability, and the 'skills gap' in higher education [Readings Online in CANVAS]

[8] Education to employment: Designing a system that works [Readings Online in CANVAS]

[9] Contextualising and integrating into the curriculum the learning and teaching of work-ready professional graduate attributes [Readings Online in CANVAS]



BUT.....

Communication and Teamwork Skills, "The Missing Ingredient" in Graduate Employability^[11]

Graduates struggle with **communication**^[14]

Poor presentation skills and an **inability to work in teams** are among the criticisms employers have of their latest graduate hires^[15]

Employers Say Students Aren't Learning Soft Skills in College^[16]

..... AND MANY MORE !!!

[\[11\] Graduates lacking soft skills, employers warn \[Readings Online in CANVAS\]](#)

[\[14\] Covid era graduates struggle with communication, say Deloitte and PwC \[Readings Online in CANVAS\]](#)

[\[15\] Are lockdown-era graduates really struggling to adapt to the workplace? \[Readings Online in CANVAS\]](#)

[\[16\] Employers Say Students Aren't Learning Soft Skills in College \[Readings Online in CANVAS\]](#)



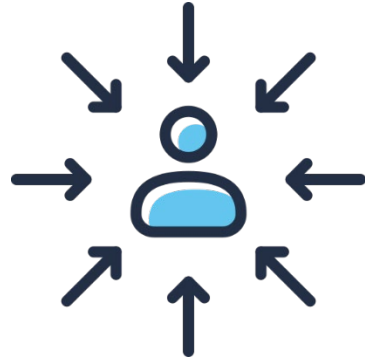
Emoji icons created by Vectors
Market - Flaticon



Communication Management



Communication Challenges



Individual icons created by Vectors Tank - Flaticon

Individual

- Semantics [meaning]
- Perception [interpretation]
- Communication Channel
- Feedback
- Anxiety
- Culture



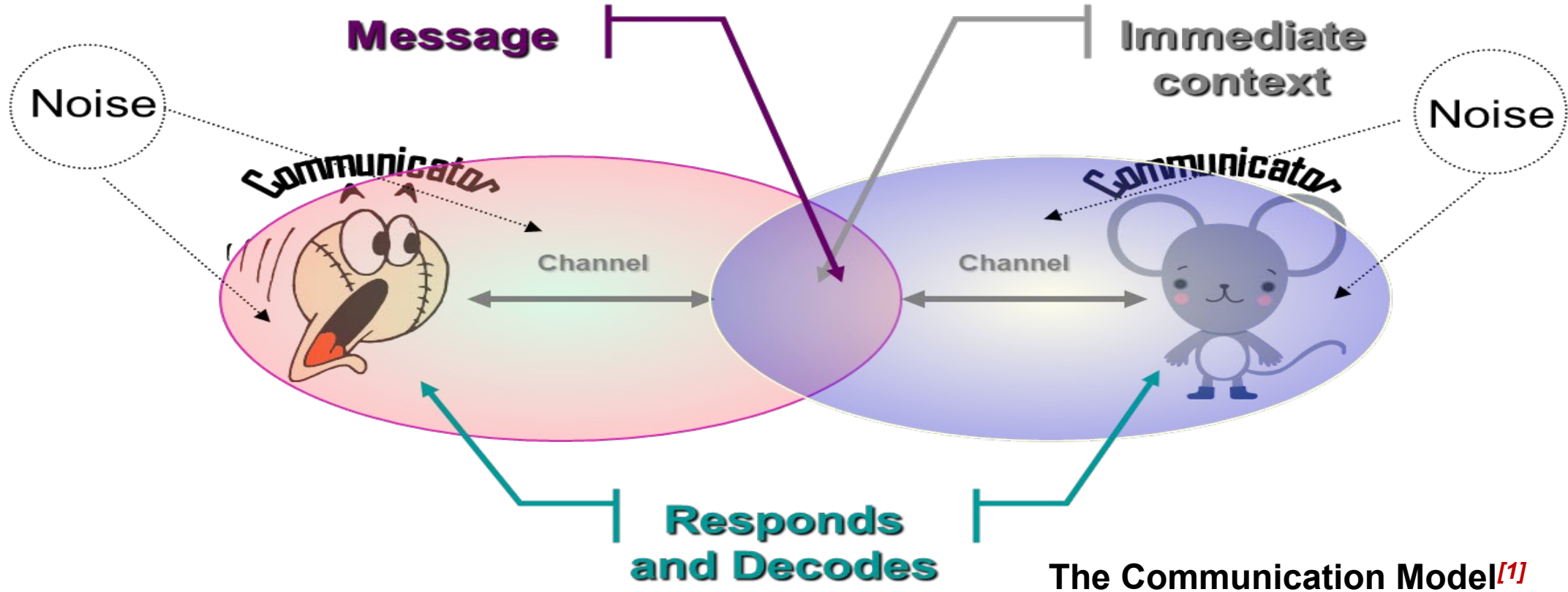
Individual icons created by Freepik - Flaticon

Team

- Status
- Silos
- Information Overload
- Lack of Communication
- Protocol [rules]

Communication Management

Communication Challenges



A mathematical theory of communication [Readings Online in CANVAS]



The importance of listening



The importance of listening



Types of Listening

Passive Listening – *Lectures*

- Taking in the information without processing or reacting

Active or Empathetic Listening – *Tutorials*

- Show interest
- Asks questions
- Avoid distractions
- Use direct eye contact
- Do not interrupt
- Read both verbal and nonverbal messages

The importance of listening



The Importance of Active Listening

Shows the speaker you are concerned or interested

Leads to getting better information

Encourages further communication

Has the potential to enhance relationships

Can calm down someone who is upset

Invites others to listen to you

Leads to better co-operation and problem solving

The importance of listening



Challenges to Listening

Physiological limitations

Inadequate background information

Selective memory or expectations

Fear of being influenced / persuaded

Bias and being judgemental

Boredom or interference from emotions

Partial listening and distractions e.g., cell phones / background noise

Physical barrier e.g., environment, lighting, uncomfortable seating

Cultural differences [understanding the spoken words]

Past experiences

Jargon & Acronyms

Communication key skills and importance



Communication Skills are critical in Project Management

Conveying your point of view

Motivating and influencing others

Delegating

Recognising, defining and solving problems

Delivering presentations / updates

Setting goals & articulating a vision

Managing conflict

Networking

Negotiating

Art of communication in project management [Readings Online in CANVAS]



L3.1 – Teams, Individuals and Motivation

So why is this important to Project Management?



SWEN90016 Software Processes and Project Management

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IT ALL STARTS HERE



Communication key skills and importance



Why Is This Important?

Because successful Project Managers MUST have the ability to:

- Read / understand the client

- Run a meeting

- Communicate (written & orally) thoughts accurately

- Manage the team

- Influence your environment

- Ensure alignment and buy-in to the purpose / outcome

Communication Plan



A large proportion of a Project Managers time is spent on communication

Project Managers often use a Communications Plan to assist in managing and coordinating key communication messages

A good project Communication Plan:

Ensures communications is effective and efficient

Allows the Project Manager to be pro-active

Sets a common understand of what will be done and when

Clarifies who is responsible for key items, what will be delivered and by who

[Communications management plan Template \[Project Management Docs\]](#)



Communication Plan



A Communications Plan defines:

- What information will be communicated - detail and format
- Communication Channel - meetings, email, telephone, web portal, etc.
- When information will be distributed – frequency of formal and informal comms
- Who is responsible
- Communication needs of stakeholders
- Resources the project will allocate for communication
- How sensitive or confidential information will be communicated & who will authorise this
- The flow of project communications
- Any constraints (internal or external) which affect project communications
- Any standard templates, formats, or documents the project must use
- Escalation process for resolving any communication-based conflicts or issues

Key Communication Considerations



Importance of Face-to-Face meetings

55% of communication is through body language^[1]

38% of communication is through how the words are said^[1]

7% of communication is through content or words that are spoken^[1]

Pay attention to more than just the actual words

A person's tone or voice and body language can say a lot about how they really feel

Nonverbal comms govern how other people feel about us and also how we feel about ourselves

[\[1\] How Much of Communication Is Nonverbal? \[Readings Online in CANVAS\]](#)

[Body Language- Ted Talk](#)

[Strong Nonverbal Skills Matter Now More Than Ever In This "New Normal" \[Readings Online in CANVAS\]](#)



Key Communication Considerations



Items to remember

- Rarely does the receiver interpret a message exactly as the sender intended
- Geographical location and cultural background affect the complexity of communications
 - Different working hours
 - Language barriers
 - Different cultural norms
- Communication helps manage conflicts effectively
- Spend time developing communication skills – *practice & feedback*
- Choose the channel appropriately

Key Communication Considerations

Communication Channels – some examples

Choose carefully as it will make a difference

1 = Most suited, 2 = Less suited and 3 = Least Suited

How well medium is Suited to:	Hard Copy	Telephone Call	Voice Mail	eMail	Meeting / f2f	Web Site
Confirming commitments	1	3	3	1	2	1
Building consensus	3	2	3	3	1	3
Mediating conflict	3	2	3	3	1	3
Resolving misunderstanding	3	1	3	3	1	3
Addressing negative behaviour	3	2	3	2	1	3
Expressing support / appreciation	1	1	2	1	1	1
Encouraging creative thinking	2	3	3	2	1	3

[Communication Channels \[Readings Online in CANVAS\]](#)



Key Communication Considerations



Communication Conflicts

Conflict is the single most undermanaged activity in projects and if left unresolved will destroy a project. Key causes include:

- Schedule
- Intellectual disagreements
- Personalities
- Project Priorities
- Manpower
- Technical
- Administration
- Personality
- Cost

[Managing conflict as a project manager \[Readings Online in CANVAS\]](#)

[Understanding and managing conflict in a project environment \[Readings Online in CANVAS\]](#)



Please ask your questions about the SWEN90016 Lecture

Nobody has responded yet.

Hang tight! Responses are coming in.

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